

Revised Generation Interconnection System Impact Study Report

For Queue Project Q17-0750

Wrangell 100 kV
December 29, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0750. The Interconnection Customer proposes a 49.0 MW solar photovoltaic facility interconnecting at the Wrangell 100 kV substation in the Indiana area. The request was received on February 27, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 253 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,303,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
LIGHTHOUSE HILL to AES BEAVER VLY PARTNERS BEAVER VALLEY 345 KV	HILLTOP to PINEY CRK PROJECT 500 KV Line	108.4	1,193	7.3
STEWART MOUNTAIN DAM SUBSTATION to LIGHTHOUSE HILL 345 KV	HILLTOP to PINEY CRK PROJECT 500 KV L	107.8	1,059	7.1
OMNI-HACKBERRY SUBSTATION 500/230 kV Autotransformer	OMNI-HACKBERRY to NORWICH 500 KV	106.8	942	6.9
MONTPELIER ELECTRIC GENERATING STA 500 kV/230 kV Auto	MONTPELIER ELECTRIC GENERATING STATION to WABASH 500 KV line	103.4	804	18.6
TREGO to BEAVER 500 KV	ADAMS to ONANCOCK DIESEL GENERATION FACILITY 230 KV	102.7	3,123	28.0
BEAVER 230/1kV Auto	BEAVER to EUFAULA 500 KV Line	102.4	1,793	13.2
BEAVER 500/1kV Transformer Bank	BEAVER to EUFAULA 500 KV	102.4	1,793	13.2
NEWPORT TIE to CANAL RD 230 KV L	BEAVER to TREGO 500 KV	101.3	713	9.9
EUFAULA 230 kV/1 kV Autotransformer	IMPERIAL to EUFAULA SUBSTATION 500 KV Line	100.5	1,453	11.5
EUFAULA 500-1 kV Transformer	IMPERIAL to EUFAULA 500 KV	100.5	1,453	11.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
LIGHTHOUSE HILL to AES BEAVER VLY PARTNERS BEAVER VLY 345 KV	Reconductor line	\$4,491,000	0% (\$21,000)
STEWART MOUNTAIN DAM SUBSTATION to LIGHTHOUSE HILL 345 KV	Reconductor line	\$5,675,000	0% (\$26,000)
OMNI-HACKBERRY SUBSTATION 500/230 kV Autotransformer	Replace transformer with higher-rated unit	\$30,993,000	0% (\$142,000)
MONTPELIER ELECTRIC GENERATING STA 500 kV/230 kV XFMR	Replace transformer with higher-rated unit	\$22,103,000	1% (\$172,000)
BEAVER to TREGO 500 KV L	Reconductor line	\$107,655,000	0% (\$502,000)
BEAVER 230-1 kV Auto	Replace transformer with higher-rated unit	\$7,870,000	0% (\$38,000)
BEAVER SUBSTATION 500-1 kV TX	Replace transformer with higher-rated unit	\$32,481,000	0% (\$158,000)
CANAL RD to NEWPORT TIE 230 KV	Reconductor line	\$8,092,000	0% (\$39,000)
EUFAULA 230 kV/1 kV Autotransformer	Replace transformer with higher-rated unit	\$11,398,000	0% (\$55,000)
EUFAULA 500-1 kV Transformer	Replace transformer with higher-rated unit	\$31,252,000	0% (\$150,000)
	Total allocated to project		\$1,303,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.